

Programming Languages - Combined Basic Q&A;

Mixed-topic test document for checking whether the RAG system retrieves the correct programming-language section.

Python Programming - Basic Q&A;

Q1. What is Python?

Answer: Python is a high-level, interpreted, general-purpose programming language known for readable syntax and a large standard library.

Q2. What is a variable in Python?

Answer: A variable is a name that refers to a value or object. Python does not require an explicit type declaration for variables.

Q3. What is a list in Python?

Answer: A list is an ordered, mutable collection that can contain multiple values. Example: numbers = [10, 20, 30].

Q4. What is a tuple?

Answer: A tuple is an ordered collection that is generally immutable after creation. Example: point = (10, 20).

Q5. What is a dictionary?

Answer: A dictionary stores data as key-value pairs. Example: student = {'name': 'Arun', 'age': 21}.

Q6. What is a function?

Answer: A function is a reusable block of code designed to perform a particular task. It is defined using the def keyword.

Q7. What is indentation in Python?

Answer: Indentation is whitespace used to define blocks of code in Python.

Q8. What is a class?

Answer: A class is a blueprint for creating objects. It can define attributes and methods.

Q9. What is inheritance?

Answer: Inheritance allows a class to derive attributes and methods from another class, supporting code reuse.

Q10. What is an exception?

Answer: An exception is an event that disrupts normal program execution. Python can handle exceptions using try and except.

Java Programming - Basic Q&A;

Q1. What is Java?

Answer: Java is a high-level, object-oriented programming language designed to be portable across platforms through the Java Virtual Machine.

Q2. What is the JVM?

Answer: JVM stands for Java Virtual Machine. It executes Java bytecode.

Q3. What is a class in Java?

Answer: A class is a blueprint that defines fields and methods used to create objects.

Q4. What is an object?

Answer: An object is an instance of a class. It represents an entity with state and behavior.

Q5. What is inheritance in Java?

Answer: Inheritance allows one class to acquire fields and methods from another class, commonly using the extends keyword.

Q6. What is method overloading?

Answer: Method overloading means defining multiple methods with the same name but different parameter lists.

Q7. What is method overriding?

Answer: Method overriding occurs when a subclass provides its own implementation of an inherited method.

Q8. What is an interface?

Answer: An interface defines a contract that implementing classes can follow.

Q9. What is a constructor?

Answer: A constructor initializes an object when it is created and has the same name as its class.

Q10. What is an exception in Java?

Answer: An exception is an event that interrupts normal execution. Java provides try, catch, finally, throw, and throws.

SQL and Database - Basic Q&A;

Q1. What is SQL?

Answer: SQL stands for Structured Query Language and is used to manage data in relational databases.

Q2. What is a database?

Answer: A database is an organized collection of data that can be stored, accessed, and managed.

Q3. What is a table?

Answer: A table is a database structure consisting of rows and columns used to store related data.

Q4. What is a primary key?

Answer: A primary key uniquely identifies each row in a table.

Q5. What is a foreign key?

Answer: A foreign key references a key in another table and helps establish relationships between tables.

Q6. What does *SELECT* do?

Answer: The SELECT statement retrieves data from one or more tables.

Q7. What does *WHERE* do?

Answer: The WHERE clause filters rows according to a specified condition.

Q8. What is a *JOIN*?

Answer: A JOIN combines related rows from two or more tables using a matching condition.

Q9. What is *normalization*?

Answer: Normalization organizes database data to reduce unnecessary redundancy and update anomalies.

Q10. What is *GROUP BY*?

Answer: GROUP BY groups rows with the same values and is commonly used with aggregate functions.